Lista 1 MLG Lara, Marcelo

1a) $f(y;\mu) = \frac{4(y+1)\mu^y}{(\mu+2)^{y+2}} = \exp\{\log 4 + \log y+1 + y\log\mu$

$-(y+2)\log\mu+2\}$

$= \exp\{y\log\mu - y\log(\mu+2) - 2\log(\mu+2) + \log 4 + \log(y+1)\}$

$= \exp\{y\log(\frac{\mu}{\mu+2}) - 2\log\mu+2 + \log 4 + \log(y+1)\}$

$\Rightarrow \phi = 1$; $\theta = \log\frac{\mu}{\mu+2} \Rightarrow e^\theta = \frac{\mu}{\mu+2} \Rightarrow \mu = \frac{2e^\theta}{1-e^\theta}$; $b(\theta) = 2\log(\mu+2)$

$\Rightarrow b(\theta) = 2\cdot\log\left(\frac{2e^\theta}{1-e^\theta}+2\right) = 2\log 2 - 2\log(1-e^\theta)$; $\rightarrow 1-e^\theta = 2/\mu+2$

*Note: $\mu > 0 \Rightarrow \theta < 0$

$c(y,\phi) = \log 4 + \log(y+1)$ Identificamos $\theta, b(\theta), c(y;\phi)$ então a dist de y pertence a FEEL.

b) $E(y) = b'(\theta)$ e $Var(y) = b''(\theta)\phi$

$b(\theta) = 2\log 2 - 2\log(1-e^\theta) \Rightarrow \frac{db(\theta)}{d\theta} = \frac{2e^\theta}{1-e^\theta} = \frac{2\,\mu/\mu+2}{2/\mu+2} = \mu$

$b''(\theta) = \frac{2e^\theta}{(1-e^\theta)^2} = \frac{2\,\mu/\mu+2}{(2/\mu+2)^2} = \frac{\mu}{2}\cdot(\mu+2) \Rightarrow E(y) = \mu$; $Var(y) = \frac{\mu(\mu+2)}{2}$

c) Usando $D(y,\hat{\mu}) = 2\phi\{\Sigma\{y_i(\tilde{\theta}_i - \hat{\theta}_i) + b(\hat{\theta}_i) - b(\tilde{\theta}_i)\}\}$:

Temos $D(y,\hat{\mu}) = 2\cdot\left\{\Sigma\, y_i\left(\log\frac{\tilde{\mu}_i}{\tilde{\mu}_i+2} - \log\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right) + 2\log(\hat{\mu}_i+2) - 2\log(\tilde{\mu}_i+2)\right\}$

* $\theta = \log\frac{\mu}{\mu+2}$ $\quad b(\theta) = 2\log(\mu+2)$; seja $\tilde{\mu}_i = y_i$

$$= 2\cdot\left\{\Sigma\left[y_i\left(\log\frac{y_i\cdot(\hat{\mu}+2)}{\hat{\mu}_i\cdot(y_i+2)}\right) + 2\cdot\log\frac{\hat{\mu}_i+2}{y_i+2}\right]\right\}.$$

2. a) $f(y;p) = p^y(1-p)^{1-y} = \exp\{y\log p + (1-y)\log(1-p)\}$

$$= \exp\{y(\log p - \log 1-p) + \log 1-p\} = \exp\left\{y\log\frac{p}{1-p} + \log 1-p\right\}$$

$\theta = \log\frac{p}{1-p} \Rightarrow p = \frac{e^\theta}{1+e^\theta}$; $\phi = 1$; $b(\theta) = \log 1-p = -\log\frac{1}{1+e^\theta} = \log 1+e^\theta$

$c(y;\phi) = 0$ Identificamos θ, $b(\theta)$, $c(y,\phi)$ e ϕ, a dist de y (Bernoulli) de fato pertence a FEL

b) $E(Y) = b'(\theta)$ e $Var(Y) = b''(\theta)\cdot\phi$; $b(\theta) = \log 1+e^\theta$

$\Rightarrow \frac{db(\theta)}{d\theta} = \frac{e^\theta}{1+e^\theta} = p$; $\frac{d^2b(\theta)}{d\theta} = \frac{e^\theta}{(1+e^\theta)^2} = p\cdot(1-p)$ $\Rightarrow$ $E(Y) = p$

$Var(Y) = p(1-p)$

c) Usando $D(y, \hat{\mu}) = 2\phi \cdot \sum [y_i(\tilde{\theta}_i - \hat{\theta}_i) + b(\hat{\theta}_i) - b(\tilde{\theta}_i)]$:

$\mu = p$

$\phi = 1$; $\theta = \log \frac{p}{1-p}$, $b(\theta) = -\log 1-p$;

$$D(\underset{\sim}{y}, \underset{\sim}{\hat{p}}) = 2 \cdot \sum \left[y_i \left(\log \frac{y_i}{1-y_i} - \log \frac{\hat{p}_i}{1-\hat{p}_i} \right) - \log 1-\hat{p}_i + \log 1-y_i \right]$$

$$= 2 \cdot \sum \left[y_i \cdot \log \left(\frac{y_i}{\hat{p}_i} \cdot \frac{1-\hat{p}_i}{1-y_i} \right) + \log \frac{1-y_i}{1-\hat{p}_i} \right]$$

$$= 2 \cdot \sum \left[y_i \cdot \log \frac{y_i}{\hat{p}_i} + y \cdot \log \frac{1-\hat{p}_i}{1-y_i} + \log \frac{1-y_i}{1-\hat{p}_i} \right]$$

$$= 2 \cdot \sum \left[y_i \log \frac{y_i}{p_i} + (1-y_i) \cdot \log \frac{1-y_i}{1-\hat{p}_i} \right]$$

* convenção: $0 \cdot \log \frac{0}{a} = 0$

note que $Y \in \{0,1\}$, $\Rightarrow d_i = \begin{cases} -2 \log \hat{p}_i , & \text{se } y_i = 0 \\ -2 \log (1-\mu_i), & y_i = 0 \end{cases}$

3. $f(y;\rho) = \frac{\rho^y}{y(-\log 1-\rho)}$; $y = 1, 2, \dots$, $0 < \rho < 1$

$\Rightarrow f(y;\rho) = \exp\{ y \log \rho - \log y - \log(-\log 1-\rho) \}$

$= \exp\{ y \log \rho - \log(-\log 1-\rho) - \log y \}$

$\theta = \log p \Rightarrow p = e^{\theta}$; $b(\theta) = \log - \log(1-p) = \log(-\log 1 - e^{\theta})$;

$c(y, \phi) = -\log y$; $\phi = 1$. *Note: $0 < p < 1 \Rightarrow \theta < 0$

$\Rightarrow$ A dist de Y pertence a FEL.

$\mu = E(y) = b'(\theta)$ e $Var(y) = b''(\theta) \cdot \phi$ $b(\theta) = \log(-\log(1-e^{\theta}))$

$$\Rightarrow b'(\theta) = \frac{1}{-\log(1-e^{\theta})} \cdot (-1) \cdot (-e^{\theta}) = \frac{-e^{\theta}}{(1-e^{\theta}) \cdot \log(1-e^{\theta})} = \frac{-p}{(1-p)\cdot \log(1-p)}$$

$$b''(\theta) = \frac{d\,b'(\theta)}{d\theta} \quad \text{seja} \quad L(e^{\theta}) = \log(1-e^{\theta}) \quad L'(p) = \frac{-1}{1-p}$$

$$\Rightarrow b''(\theta) = \frac{d}{d\theta}\left(\frac{-e^{\theta}}{(1-e^{\theta}) \cdot L(e^{\theta})}\right) = \frac{-e^{\theta}\left[(1-e^{\theta}) \cdot L(e^{\theta})\right] + e^{\theta}\left[(-e^{\theta}) \cdot L(e^{\theta}) + (1-e^{\theta}) \cdot L'(e^{\theta}) e^{\theta}\right]}{\left[(1-e^{\theta}) \cdot L(e^{\theta})\right]^2}$$

$$= \frac{-e^{\theta} L(e^{\theta}) + e^{2\theta} L(e^{\theta}) + -e^{2\theta} L(e^{\theta}) + e^{2\theta} L'(e^{\theta}) - e^{3\theta} L'(e^{\theta})}{(1-e^{\theta})^2 \cdot L(e^{\theta})^2}$$

$$= \frac{-e^{\theta} L(e^{\theta}) + e^{2\theta} L'(e^{\theta}) - e^{3\theta} L'(e^{\theta})}{(1-e^{\theta})^2 \cdot L(e^{\theta})^2} = \frac{-p L(p) + p^2 (L(e^{\theta})) - p^3 L'(e^{\theta})}{(1-p)^2 \cdot L(p)^2}$$

$$= \frac{-p \log(1-p) + p^2\left(\frac{-1}{1-p} + \frac{p}{1-p}\right)}{(1-p)^2 \cdot L(p)^2} = \frac{-p \log(1-p) - p^2}{\left[(1-p) \log(1-p)\right]^2}$$

$D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}})$: Sabemos que $f(y;\rho) = \exp\{y_i \log\rho_i - \log L(\rho_i) - \log y_i\}$

$\Rightarrow l(\underset{\sim}{\rho}, \underset{\sim}{y}) = \sum[y_i \log\rho_i - \log L(\rho_i) - \log y_i]$

usando $D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}}) = 2\cdot[l(\underset{\sim}{y}, \underset{\sim}{y}) - l(\underset{\sim}{\hat{\rho}}, \underset{\sim}{y})]$ * $\tilde{\rho}_i = y_i$

Temos

$$2\cdot\sum[y_i \log y_i - \log L(y_i) - \cancel{\log y_i} - y_i \log\hat{\rho}_i - \log L(\hat{\rho}_i) - \cancel{\log y_i}]$$

$$= 2\cdot\sum\left[y_i \log\frac{y_i}{\hat{\rho}_i} - \log\left(\frac{L(y_i)}{L(\hat{\rho}_i)}\right)\right] = 2\cdot\sum\left[y_i \log\frac{y_i}{\hat{e}_i} - \log\left(\frac{\log(1-y_i)}{\log(1-\hat{\rho}_i)}\right)\right]$$

$$\Rightarrow D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}}) = 2\cdot\sum\left[y_i \log\frac{y_i}{\hat{\rho}_i} - \log\left(\frac{\log(1-y_i)}{\log(1-\hat{\rho}_i)}\right)\right]$$

obs: Se $y_i = 1$

$d_i = 2\cdot\log\left[\frac{\log(1-\hat{\rho}_i)}{\hat{\rho}_i}\right]$

R^2: $R^2 = 1 - \frac{D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}})}{D(\underset{\sim}{y}, \underset{\sim}{\bar{y}})} = 1 - \frac{2\sum\left[y_i \log\frac{y_i}{\hat{\rho}_i} - \log\left(\frac{L(y_i)}{L(\hat{\rho}_i)}\right)\right]}{2\cdot\sum\left[y_i \log\frac{y_i}{\bar{y}_i} - \log\left(\frac{L(y_i)}{L(\bar{y}_i)}\right)\right]}$

4. $g(\mu_i) = -\log(-\log(\mu_i))$

a) $\eta_i = -\log(-\log(\mu_i)) \Rightarrow e^{-\eta_i} = -\log\mu_i \Rightarrow e^{-e^{-\eta_i}} = \mu_i$

$\mu_i = e^{-e^{-\eta_i}}$ //

b) note que $e^{-n_i} > 0 \Rightarrow -e^{-n_i} < 0 \Rightarrow e^{-e^{-n_i}} < 1$

e como $e^x > 0 \ \forall x \in \mathbb{R}$, então $0 < e^{-e^{-n_i}} < 1$.

5. $\underset{\sim}{z}$ e W em função de μ_i. Goma.

a) g canônica $\eta_i = g(\mu_i) = 1/\mu_i \Rightarrow g'(\mu_i) = -1/\mu_i^2$

$V(\mu_i) = \mu_i^2$.

Sabe-se que $\underset{\sim}{z} = \underset{\sim}{\eta} + G \cdot (\underset{\sim}{y} - \underset{\sim}{\mu})$ e $w_i = \frac{1}{V_i}\left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2 = \frac{1}{V_i}\left[\left(\frac{\partial \eta_i}{\partial \mu_i}\right)^{-1}\right]^2$

$\Rightarrow z_i = \eta_i + (y_i - \mu_i) \cdot g'(\mu_i) = \frac{1}{\mu_i} + (y_i - \mu_i) \cdot \frac{(-1)}{\mu_i^2} = \frac{1}{\mu_i} - \frac{(y_i - \mu_i)}{\mu_i^2}$

$= \frac{2}{\mu_i} - \frac{y_i}{\mu_i^2}$; $w_i = \frac{1}{\mu_i^2}\left[\left(\frac{-1}{\mu_i^2}\right)^{-1}\right]^2 = \mu_i^2$

$\Rightarrow \underset{\sim}{z} = \begin{bmatrix} 2/\mu_1 - y_1/\mu_1^2 \\ \vdots \\ 2/\mu_n - y_n/\mu_n^2 \end{bmatrix}$ e $W = \text{diag}\{\mu_1^2, \dots, \mu_n^2\}$

b) g log $\quad \eta_i = g(\mu_i) = \log \mu_i \Rightarrow g'(\mu_i) = 1/\mu_i \quad V(\mu_i) = \mu_i^2$

$z_i = \eta_i + (y_i - \mu_i) \cdot g'(\mu_i) = \log \mu_i + (y_i - \mu_i)/\mu_i = \log \mu_i + \frac{y_i}{\mu_i} - 1$

$$w_i = \frac{1}{V_i}\left(\left[g'(\mu_i)\right]^{-1}\right)^2 = \frac{1}{\mu_i^2}\left[\left(\frac{1}{\mu_i}\right)^{-1}\right]^2 = \frac{\mu_i^2}{\mu_i^2} = 1 \Rightarrow W = I_{n\times n}$$

$$\underset{\sim}{z} = \begin{bmatrix} \log\mu_1 + y_1/\mu_1 - 1 \\ \vdots \\ \log\mu_n + y_n/\mu_n - 1 \end{bmatrix} \quad e \quad W = I_{n\times n} \quad "$$

6. mostre $\hat{\phi}$ por MV é $n/D^*(y;\hat{\mu})$. Y normal

$Y \sim N(\mu, \sigma^2)$ $\quad Var(Y) = \sigma^2 = \phi^{-1}$

$$\Rightarrow L(\underset{\sim}{\mu}, \phi; \underset{\sim}{y}) = \prod_{i=1}^{n} \frac{\sqrt{\phi}}{\sqrt{2\pi}} \exp\left\{-\frac{\phi}{2}(y_i-\mu_i)^2\right\} = \left(\frac{\phi}{2\pi}\right)^{\frac{n}{2}} \cdot \exp\left\{\sum_{i=1}^{n} -\frac{\phi}{2}(y_i-\mu_i)^2\right\}$$

$$\Rightarrow \ell(\underset{\sim}{\mu}, \phi, \underset{\sim}{y}) = \frac{n}{2}\log\phi - \frac{n}{2}\log 2\pi - \frac{\phi}{2}\sum(y_i-\mu_i)^2$$

$$\Rightarrow \frac{d\ell}{d\phi} = \frac{n}{2}\cdot\frac{1}{\phi} - \frac{\sum(y_i-\hat{\mu}_i)^2}{2} \; ; \; \left.\frac{d\ell}{d\phi}\right|_{\phi=\hat{\phi}} = 0 \Rightarrow \frac{n}{2\hat{\phi}} = \frac{\sum(y_i-\hat{\mu}_i)^2}{2}$$

$$\Rightarrow \hat{\phi}_{MV} = \frac{n}{\sum(y_i-\mu_i)^2} \; ; \; {}^{*}\frac{d^2\ell}{d\phi^2} < 0 \; ; \text{ note agora } D^*(y;\hat{\mu}) =$$

$$D^*(\underset{\sim}{y}\,\underset{\sim}{\hat{\mu}}) = 2\cdot\sum\left[y_i(y_i-\hat{\mu}_i) + \hat{\mu}_i^2/2 - y_i^2/2\right] = \sum(y_i-\hat{\mu}_i)^2$$

* pois $\theta_i = y_i \Rightarrow \tilde{\theta} = y_i$ e $\hat{\theta} = \mu_i$; e $b(\theta) = \frac{\theta_i^2}{2}$; $\frac{\hat{\mu}_i^2}{2} - \frac{y_i^2}{2} + y_i^2 - y_i\hat{\mu}_i = \frac{(y_i^2 - 2y_i\hat{\mu}_i + \mu_i^2)}{2}$

$$\Rightarrow \hat{\phi} = \frac{n}{D^*(\underset{\sim}{y}; \hat{\underset{\sim}{\mu}})}$$ ■

7. $Y \sim$ Gaussiana Inversa . g canonica $\underset{\sim}{y} = [2,3; 1,8; 1,5; 1,3; 1,2]^T$

$\underset{\sim}{x} = [2; 4; 6; 9; 10]^T$

β_0 e β_1 por MV

ϕ por MV e por MM

Estimação de $\underset{\sim}{\beta}$: $\quad V(\mu_i) = \mu_i^3$

g canônica: $\eta_i = g(\mu_i) = \frac{1}{\mu_i^2} = \mu_i^{-2} \Rightarrow \mu_i = (\eta_i)^{-\frac{1}{2}}$; $g'(\mu_i) = -2\mu^{-3}$

$$w_i = \frac{1}{V_i}\left(g'(\mu_i)^{-1}\right)^2 = \frac{1}{\mu_i^3} \cdot \left[-\frac{\mu^3}{2}\right]^2 = \frac{\mu_i^3}{4};$$

$$z_i = \eta_i + (y_i - \mu_i)\, g'(\mu_i) = \frac{1}{\mu_i^2} + (y_i - \mu_i)\cdot(-2\mu_i^{-3}) = \frac{1}{\mu_i^2} - \frac{(y_i - \mu_i)}{\mu_i^3}$$

Inicializando com $\mu^{(0)} = y_i \Rightarrow W^{(0)} = \text{diag}\left\{\frac{y_1^3}{4}; \frac{y_2^3}{4}; \dots; \frac{y_5^3}{4}\right\}$

e

$$\underset{\sim}{z}^{(0)} = [1/y_1^2; 1/y_2^2; \dots; 1/y_5^2]^T$$

Temos $\underset{\sim}{\beta}^{(1)} = (X^T W^{(0)} X)^{-1} X^T W^{(0)} \underset{\sim}{z}^{(0)}$, onde $X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 9 & 10 \end{bmatrix}^T$

e $X^T W^{(0)} X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 9 & 10 \end{bmatrix} \frac{1}{4} \begin{bmatrix} 2,3^2 & & & & \\ & 1,8^2 & & & \\ & & 1,5^2 & & \\ & & & 1,3^2 & \\ & & & & 1,2^2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 9 \\ 1 & 10 \end{bmatrix}$

$= \begin{bmatrix} 6,32475 & 26,24125 \\ 26,24125 & 153,55925 \end{bmatrix} \Rightarrow (X^T W^{(0)} X)^{-1} \simeq \begin{bmatrix} 0,5433 & -0,092 \\ -0,092 & 0,223 \end{bmatrix}$

e $X^T W^{(0)} \underset{\sim}{z} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 9 & 10 \end{bmatrix} \frac{1}{4} \begin{bmatrix} 2,3^3 & & & & \\ & 1,8^3 & & & \\ & & 1,5^3 & & \\ & & & 1,3^3 & \\ & & & & 1,2^3 \end{bmatrix} \begin{bmatrix} 1/2,3^2 \\ 1/1,8^2 \\ 1/1,5^2 \\ 1/1,3^2 \\ 1/1,2^2 \end{bmatrix}$

$= \begin{bmatrix} 2,025 \\ 11,125 \end{bmatrix} \Rightarrow \underset{\sim}{\beta}^{(1)} = (X^T W^{(0)} X)^{-1} X^T W^{(0)} \underset{\sim}{z}^{(0)}$

$\underset{\sim}{\beta}^{(1)} = \begin{bmatrix} 0,5433 & -0,092 \\ -0,092 & 0,223 \end{bmatrix} \cdot \begin{bmatrix} 2,025 \\ 11,125 \end{bmatrix} = \begin{bmatrix} 0,067 31134 \\ 0,060 94499 \end{bmatrix}$

Feita a 1ª iteração, temos $\eta^{(1)} = X^T \underset{\sim}{\beta}^{(1)} \Rightarrow \mu^{(1)} = (X^T \beta^{(1)})^{\frac{-1}{2}}$

$\Rightarrow \mu^{(1)} = (X^T \underset{\sim}{\beta}^{(1)})^{\frac{1}{2}} = [2,2989 \quad 1,7929 \quad 1,5197 \quad 1,2743 \quad 1,2155]^T$

$\Rightarrow \underset{\sim}{z}^{(1)} = [0,19 \quad 0,31 \quad 0,44 \quad 0,59 \quad 0,69]^T$

$$W^{(1)} = \text{diag}\{3{,}037;\ 1{,}44;\ 0{,}87;\ 0{,}52;\ 0{,}45\}$$

$$\Rightarrow \underset{\sim}{\beta}^{(2)} = (X^T W^{(1)} X)^{-1} X^T W^{(1)} \underset{\sim}{z}^{(1)} = \begin{bmatrix} 0{,}06714 \\ 0{,}06101 \end{bmatrix}$$

note que a diferença entre $\beta^{(2)}$ e $\beta^{(1)}$ é pequena, vamos para na 2ª iteração, pois parece que convergiu $\Rightarrow \tilde{\beta} = \begin{bmatrix} 0{,}06714 \\ 0{,}06101 \end{bmatrix}$

$$\Rightarrow \underset{\sim}{\eta}^{(2)} = [0{,}1892 \quad 0{,}3112 \quad 0{,}4332 \quad 0{,}6163 \quad 0{,}6773]^T$$

$$\text{e} \quad \underset{\sim}{\mu}^{(2)} = [2{,}2992 \quad 1{,}7926 \quad 1{,}5193 \quad 1{,}2738 \quad 1{,}2151]^T$$

Estimação de ϕ por método dos momentos:

$$V(\mu) = \mu_i^3 \text{ e } Var(Y) = \frac{1}{\phi} \cdot V(\mu) \Rightarrow Var(Y) = \frac{\mu_i^3}{\phi}$$

$$\text{e } Var(Y_i) = E[(Y_i - \mu_i)^2] = \frac{1}{\phi}\mu_i^3 \Rightarrow E\left[\frac{(Y_i - \mu_i)^2}{\mu_i^3}\right] = \frac{1}{\phi}$$

$$\Rightarrow \frac{1}{\hat{\phi}} = \frac{1}{n} \cdot \sum \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i^3} \Rightarrow \hat{\phi} = \frac{n}{\sum\left[\frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i^3}\right]}$$

usando $\hat{\mu}_i = \tilde{\mu}_i^{(2)}$

$$\Rightarrow \hat{\phi}_{mm} = \frac{5}{0{,}0005740679} \cong 8709{,}771$$

Alternativa: $k=2$ e $\phi\left[\sum \frac{(y_i-\hat{\mu}_i)^2}{V(\hat{\mu}_i)}\right] \to \chi^2_{n-k-1}$

$$\Rightarrow \hat{\phi}^{\text{AJUSTADO}}_{MM} = \frac{n-k-1}{\sum \frac{(y_i-\hat{\mu}_i)^2}{\hat{\mu}_i^3}} \cong \frac{2}{0,0005740679} \cong 3493,908$$

Estimação de ϕ por MV na parametrização atual:

$$f(y_i;\mu_i,\phi) = \sqrt{\frac{\phi}{2\pi y_i^3}} \cdot \exp\left\{-\frac{\phi}{2}\frac{(y_i-\mu_i)^2}{\mu_i^2 y_i}\right\} \Rightarrow$$

$$l(\phi) = \frac{n}{2}\log\phi - \frac{\phi}{2}\sum \frac{(y_i-\mu_i)^2}{\mu_i^2 y_i} + \text{constantes p/ } \phi$$

$$\Rightarrow U_\phi = \frac{dl}{d\phi} = \frac{n}{2\phi} - \frac{1}{2}\sum \frac{(y_i-\hat{\mu}_i)^2}{\mu_i^2 y_i} \Rightarrow U_\phi = 0 \Rightarrow \hat{\phi} = \frac{n}{\sum \frac{(y_i-\hat{\mu}_i)^2}{\hat{\mu}_i^2 y_i}}$$

usando $\hat{\mu}_i = \tilde{\mu}_i^{(2)}$ temos

$$\hat{\phi}_{MV} \cong \frac{5}{0,0005703 27} \cong 8766,9 \text{ //}$$